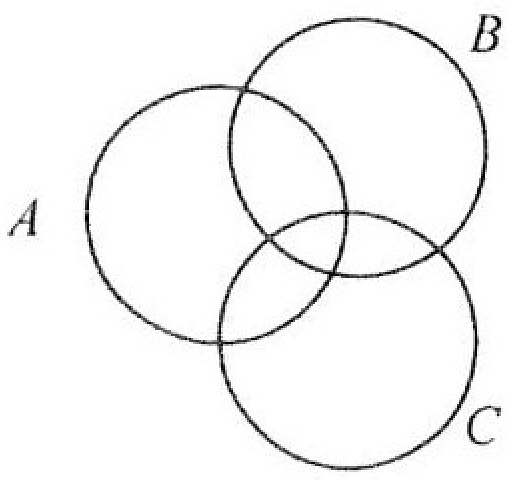
# Unit 10

## *Sets and Venn Diagrams*

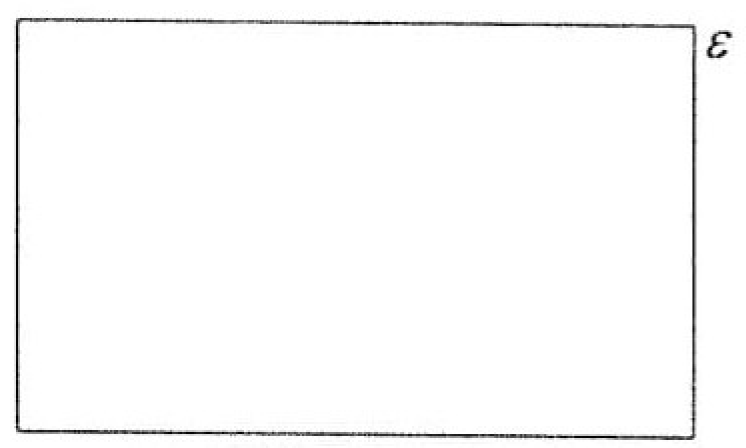
1. (a) On the Venn diagram in the answer space, shade the set $(A \cap B) \cup C$.




(b) The Universal set is the set of all positive integers and $P = \{x : x < 10\}$.

Complete the statement: $P' = \{x :$ ________________ $\}$ (J2000/1/7)

2. (a) $A$, $B$ and $C$ are subsets of the universal set $\varepsilon$, $B \cup A = B$ and $B \cap C = \Phi$. Illustrate this on the Venn diagram in the answer space.




(b) In a group of people, 15 drink tea, 28 drink coffee, 12 drink both tea and coffee and 6 drink neither tea nor coffee. By drawing a Venn diagram, or otherwise, find the total number of people in the group. (Int'l Exam N2000/1/10)

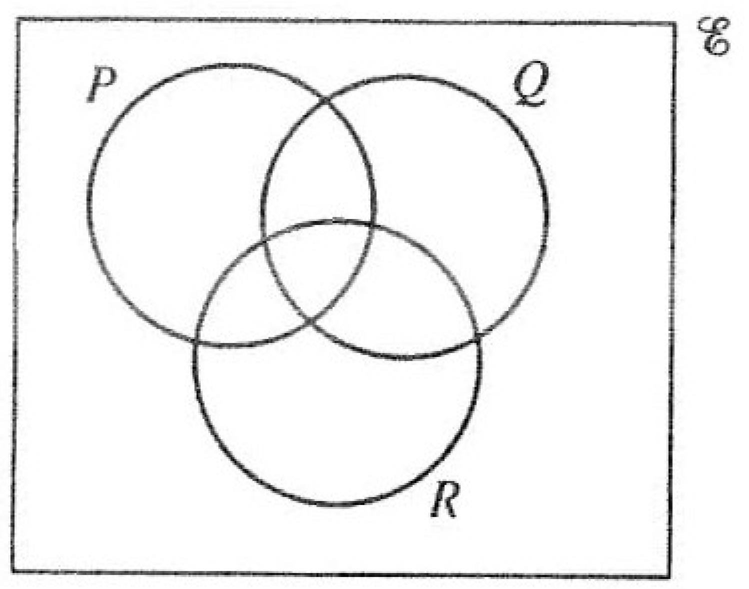
3. $\varepsilon = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
$A = \{x: x \text{ is an even number}\}$
$B = \{x: x \text{ is a multiple of } 3\}$
(a) Draw a Venn Diagram to illustrate these sets, indicating the members of each subset.
(b) (i) A number $n$ is chosen at random from $\varepsilon$. Find the probability that

(a) $n \in A$,
(b) $n \in A'$,
(c) $n \in A \cap B$,
(d) $n = 14$.

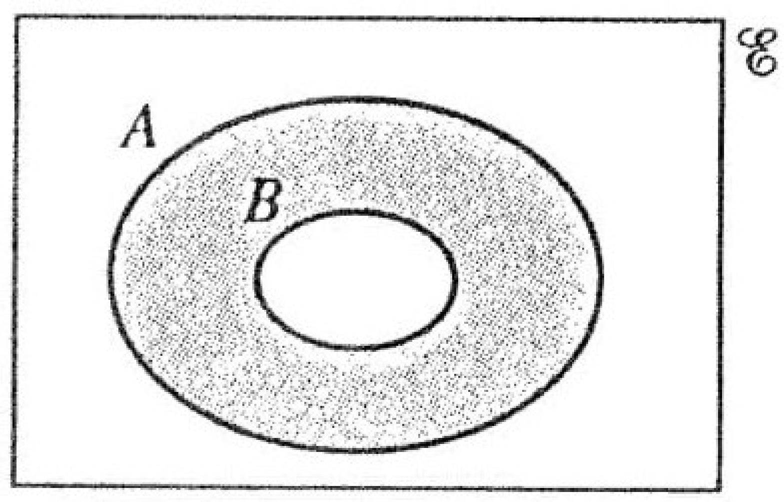
(ii) A second number $m$ is chosen at random from the **complete** set $\varepsilon$. Find the probability that the sum of $n$ and $m$ is 18. (J2001/2/4)

4. (a) On the Venn Diagram, shade the set

$$(P \cup R) \cap Q.$$



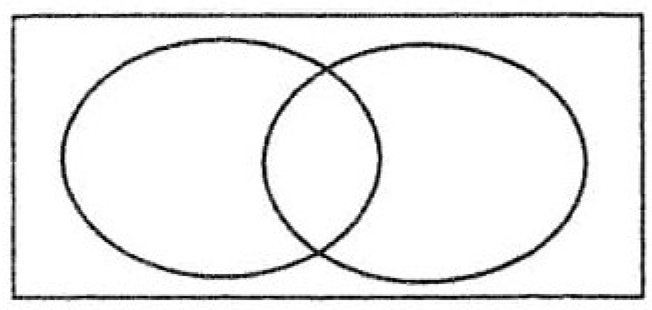

(b) Express in set notation, as simply as possible, the set shaded in the Venn Diagram.




(c) There are 27 people in a restaurant. Of these, 10 eat fish, 15 eat meat and 9 eat neither fish nor meat.
Using a Venn Diagram, or otherwise, find the number of people in the restaurant who eat fish but not meat. (Int'l Exam N2001/1/20)

5. There are 50 people on a tour. One day, 26 people went on the morning cruise and 29 to the evening barbecue. Using Venn diagrams, or otherwise, answer the following questions.

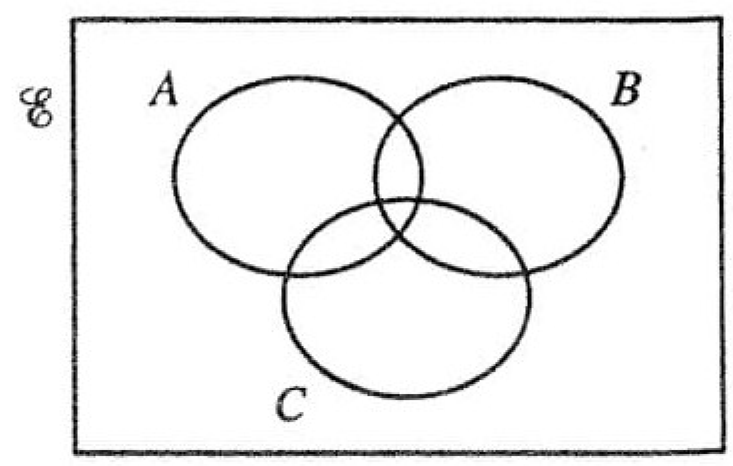
(a) It was thought that 4 people went to both events and 1 person to neither. Explain why this was not possible.



(b) Find the least number and the greatest number of people who could have gone to both events. (J2002/1/9)

6. (a) On the Venn diagram in the answer space, shade the set $(A' \cup B') \cap C$

(b) $\varepsilon = \{x : x \text{ is an integer and } 4 \leqslant x \leqslant 16\}$
$P = \{x : x \text{ is a prime number}\}$
$S = \{x : x \text{ is an odd number}\}$
$T = \{x : x \text{ is a multiple of } 5\}$
(i) List the members of the set $S \cap T$.
(ii) Describe, **in words**, the set $S'$.
(iii) Find $n(P \cup T)$.



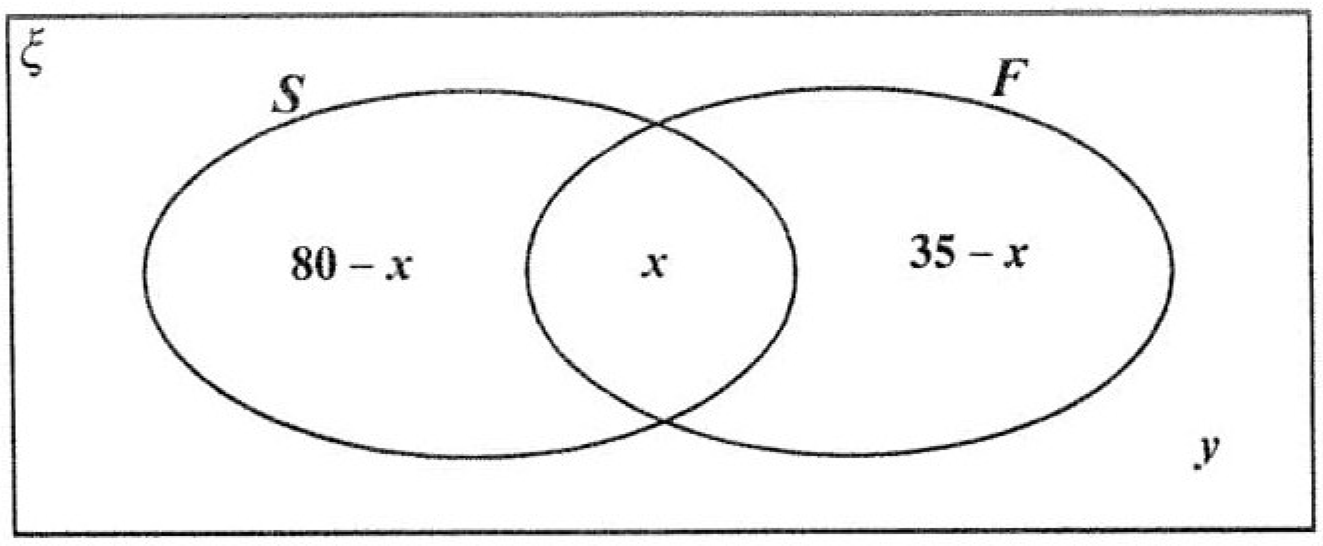

(Int'l Exam N2003/1/15)

7. In a group of $100$ students, $80$ study Spanish and $35$ study French.

$x$ students study Spanish and French.

$y$ students study neither Spanish nor French.

The Venn diagram illustrates this information.




(a) Expressed in set notation, the value of $x$ is $n(S \cap F)$.

Express the value of $y$ in set notation.

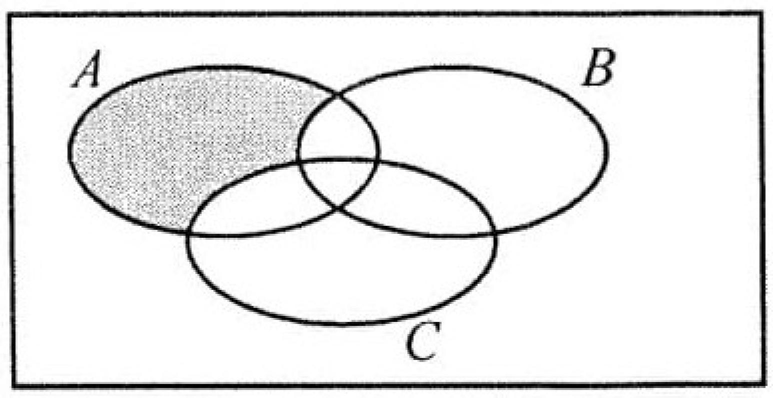
(b) Find, in its simplest form, an expression for $y$ in terms of $x$.

(c) Find

(i) the least possible value of $x$,

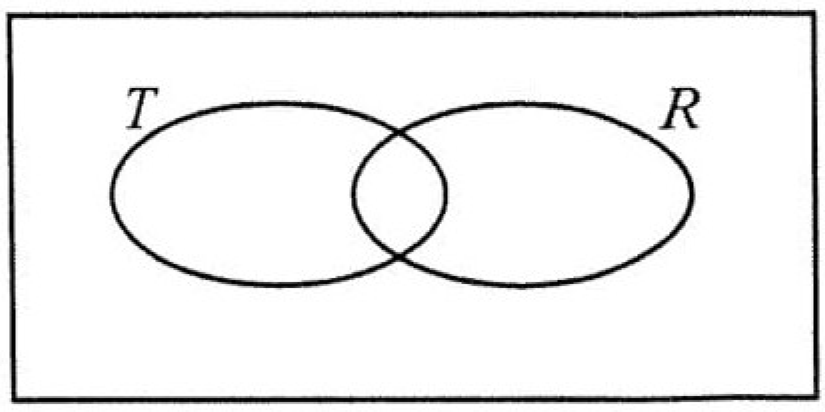
(ii) the greatest possible value of $y$. (J2004/2/5)

8. (a) Express in set notation, as simply as possible, the subset shaded in the Venn diagram.




(b) $\xi = \{$ all polygons$\}$ ,
$T = \{$ all triangles$\}$ ,
$R = \{$ all regular polygons$\}$
$Q = \{$ all quadrilaterals$\}$ .

Add the set $Q$ to the Venn diagram in the answer Space.

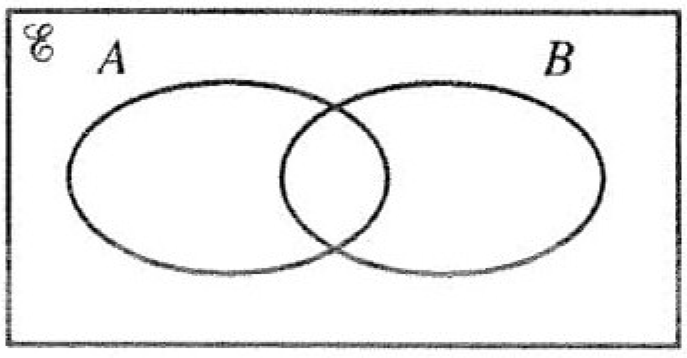



(c) $\xi = \{x : x \text{ is an integer and } 3 \leq x \leq 11\}$.
$F = \{x : x \text{ is a factor of } 12\}$,
$O = \{x : x \text{ is an odd number})$.

List the elements of the set $(F \cup O)'$ .

(d) It is given that $n(\xi) = 20$.
$P$ and $S$ are sets such that $n(P) = 7$ and $n(S) = 16$.
Find the smallest possible value of $n(P \cap S)$. (N2004/1/18)

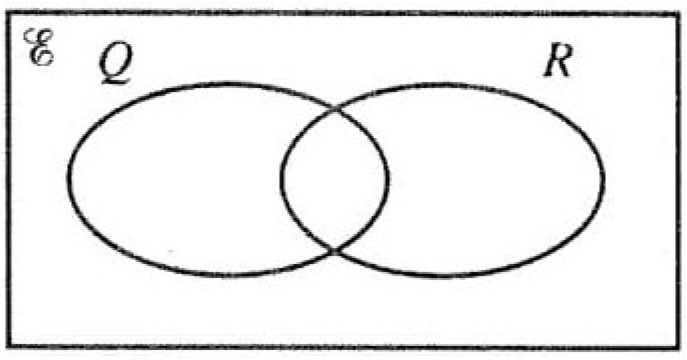
9. (a) On the Venn Diagram in the answer space, shade the region $A \cup B'$




(b) $\xi$ = {all polygons}, $Q$ = {all quadrilaterals}, $R$ = {all regular polygons}.

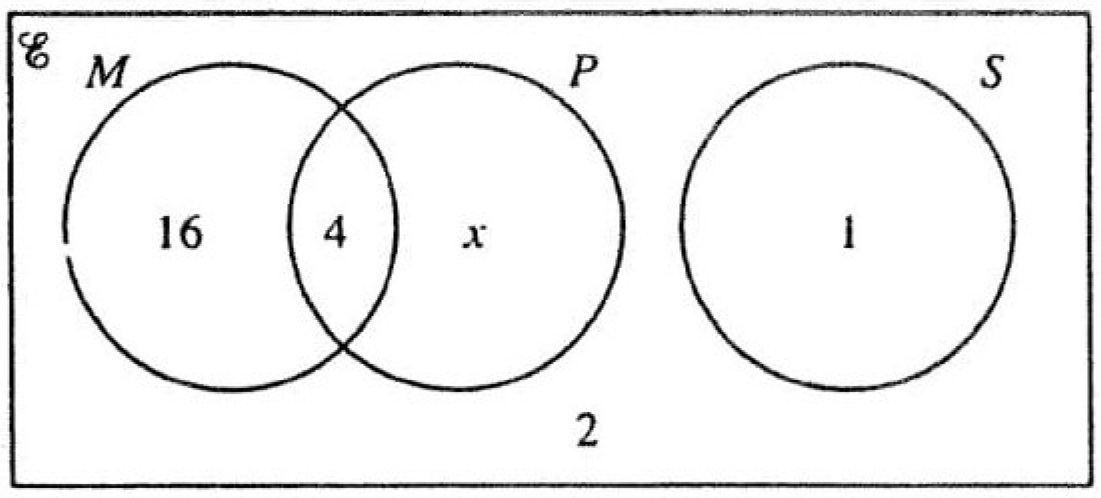
(i) What is the special name of the polygons which belong to $Q \cap R$

(ii) On the Venn Diagram in the answer space, show the set $T$ = {all equilateral triangles}.




(N2005/1/13)

10. (a) The results of a survey of 31 students are shown in the Venn diagram.
$\mathscr{E}$ = {students questioned in the survey}
$M$ = {students who study Mathematics}
$P$ = {students who study Physics}
$S$ = {students who study Spanish}

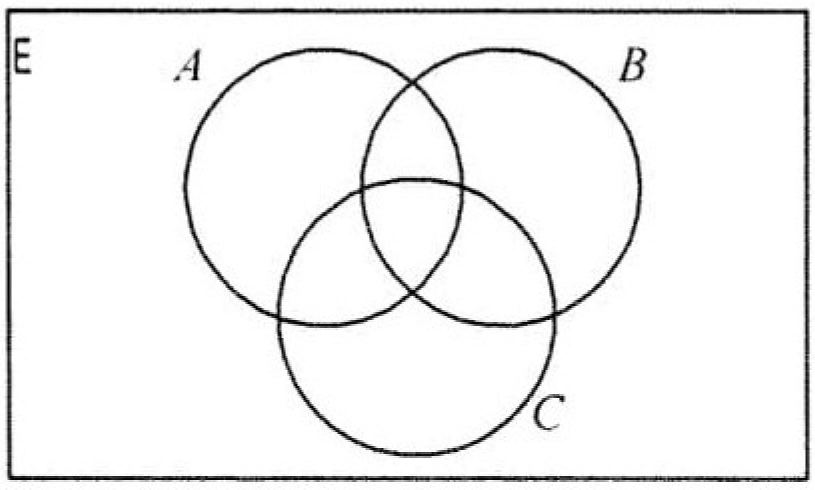



(i) Write down the value of
(a) $x$,
(b) $n(M \cap P)$,
(c) $n(M \cup S)$,
(d) $n(P')$.

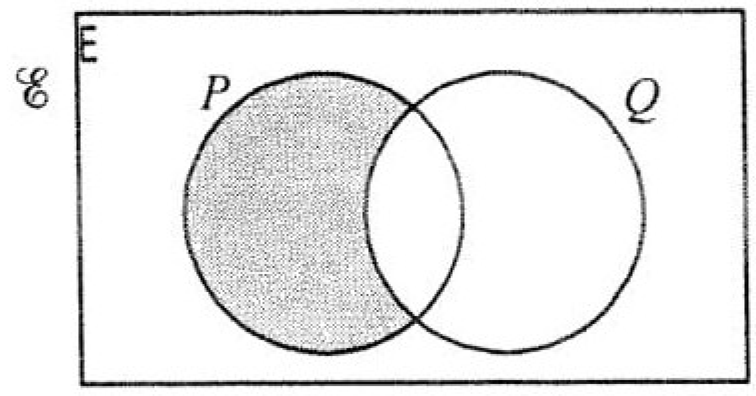
(ii) Write down a description, in words, of the set that has 16 members.

(J2006/2/6a)

11. (a) On the Venn diagram, shade the set $A \cup (B \cap C)$.

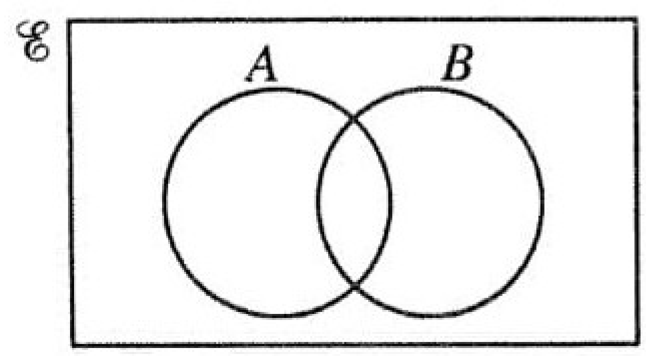


(b) Express in set notation the subset shaded in the Venn diagram.




(c) In a class of 36 students, 25 study History, 20 study Geography and 4 study neither History nor Geography.
Find how many students study both History and Geography. (N2006/1/18)

12. (a) The sets $A$ and $B$ are shown on the Venn Diagram in the answer space.
The element $y$ is such that $y \in A$ and $y \notin B$.
On the diagram, write $y$ in the correct region.




(b) $\mathscr{E} = \{x : x \text{ is an integer and } 1 \le x \le 8\}$.

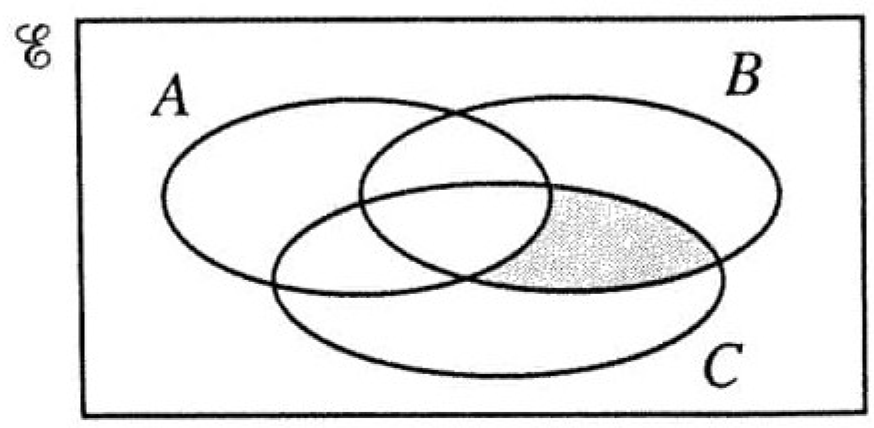
$P = \{x : x > 5\}$.

$Q = \{x : x \le 3\}$.

(i) Find the value of $n(P \cup Q)$.

(ii) List the elements of $P' \cap Q'$. (J2007/1/9)

13. (a) Express, in set notation, as simply as possible, the subset shaded in the Venn diagram.




(b) It is given that $n(\mathscr{E}) = 40$, $n(P) = 18$, $n(Q) = 20$ and $n(P \cap Q) = 7$.
Find

(i) $n(P \cup Q)$,

(ii) $n(P' \cap Q')$.

(N2007/1/9)

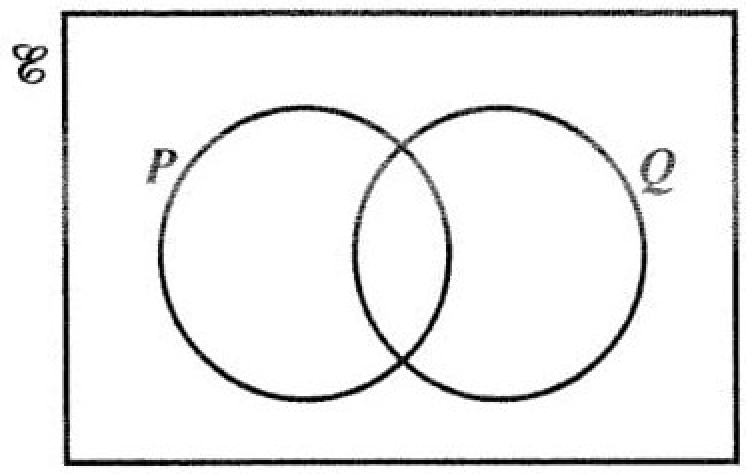
14. Mary has 50 counters.
Some of the counters are square, the remainder are round.
There are 11 square counters that are green.
There are 15 square counters that are not green.
Of the round counters, the number that are not green is double the number that are green.
By drawing a Venn diagram, or otherwise, find the number of counters that are

(i) round,

(ii) round and green,

(iii) not green. (J2008/2/5a)

15. (a) On the Venn diagram shown in the answer space, shade the set $P \cup Q'$.




(b) There are 27 children in a class.
Of these children, 19 own a bicycle, 15 own a scooter and 3 own neither a bicycle nor a scooter.
Using a Venn diagram, or otherwise, find the number of children who own a bicycle but not a scooter. (N2008/1/9)

16. (a) $\mathscr{E} = \{ 1, 2, 3, 4, 5 \}$,
$A = \{ 1, 2, 3 \}$,
$B = \{ 5 \}$,
$C = \{ 3, 4 \}$.

List the elements of

(i) $A \cup C$

(ii) $B' \cap C'$.

(b) A group of 60 children attend an after school club.
Of these, 35 children play football and 29 play hockey.
3 children do not play either football or hockey.

By drawing a Venn diagram, or otherwise, find the number of children who play only hockey. (J2009/1/18)

17. (a) $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$L = \{x : x \text{ is an odd number}\}$

$M = \{x : x \text{ is a multiple of } 3\}$

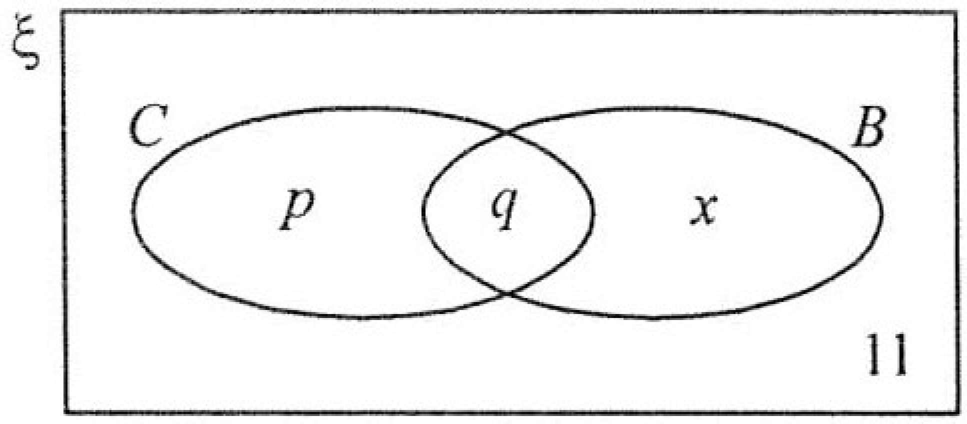
(i) Write down

(a) $L \cap M$,

(b) $L' \cap M$.

(ii) A number $n$ is chosen at random from $\xi$.
Find the probability that $n \in L \cup M$.

(b) In a survey, a number of people were asked "Do you own a car?" and "Do you own a bicycle?". The Venn diagram shows the set $C$ of car owners and the set $B$ of bicycle owners.
The letters $p$, $q$ and $x$ are the numbers of people in each subset.
11 people owned neither a car nor a bicycle.




A total of 66 people owned a car.
4 times as many people owned a car **only** as owned a bicycle **only**.

(i) Write down expressions, in terms of $x$, for

(a) $p$,

(b) $q$.

(ii) A total of 27 people owned a bicycle. Calculate

(a) $x$,

(b) the **total** number of people who were in the survey. (N2009/2/4)

18. $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$
$A = \{x : x \text{ is a multiple of } 3\}$
$B = \{x : x \text{ is a factor of } 24\}$
$C = \{x : x \text{ is an odd number}\}$

(i) Find

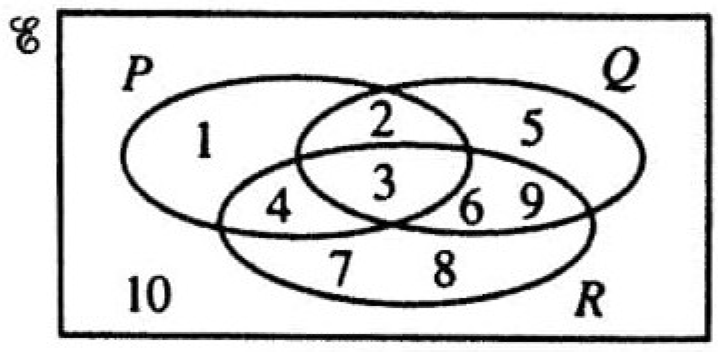
(a) $n(B)$,

(b) $(A \cup B \cup C)'$.

(ii) A number, $k$, is chosen at random from $\mathscr{E}$.
Find the probability that $k \in A \cap B$. (J2010/2/5c)

19. The Venn diagram shows the sets $\mathscr{E}$, $P$, $Q$ and $R$.

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$



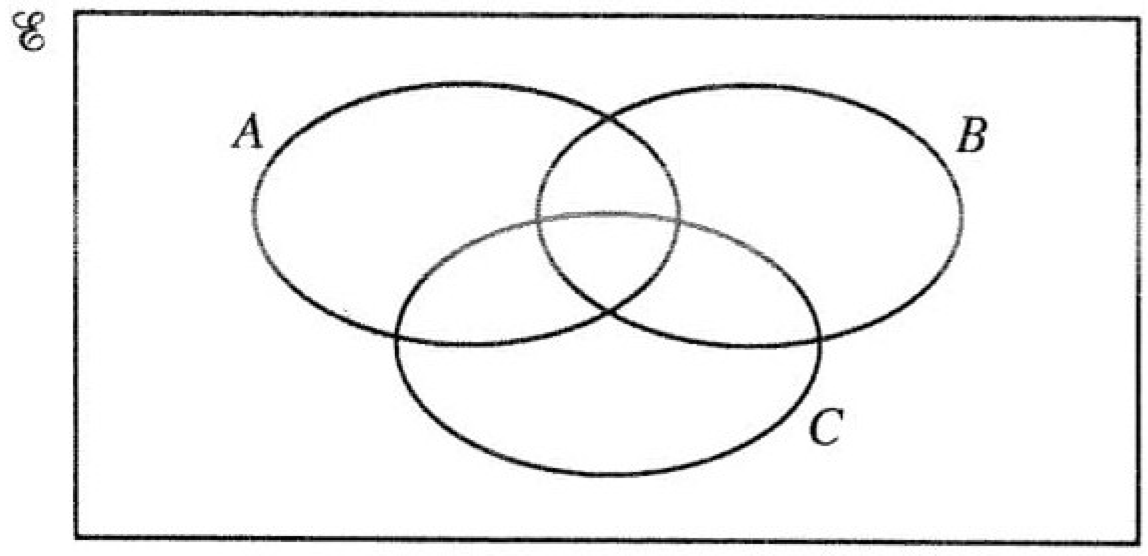

(a) Find the value of $n(Q \cup R)$.

(b) List the elements of the set $P' \cap (Q \cup R)$.

(c) One element is chosen at random from $\mathscr{E}$.
Write down the probability that this element belongs to $(P \cap Q) \cup (P \cap R)$.

(N2010/1/11)

20. (a) On the Venn diagram, shade the set $A \cap B \cap C'$.




(b) $\mathscr{E}$ = {2, 3, 4, 5, 6, 7, 8, 9, 10}
P = $\{x : x$ is a prime number$\}$
$Q = \{x : x \geq 5\}$

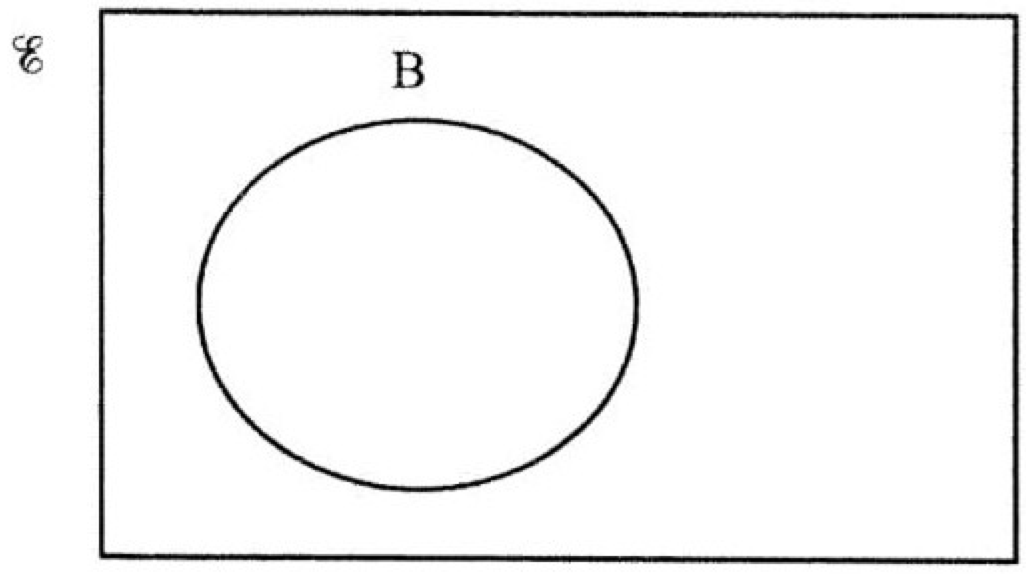
(i) Find the value of $n(P \cap Q)$.

(ii) List the elements of $P \cup Q'$. (J2011/1/12)

21. The Venn diagram shows the Universal set and the set B.
A and C are two sets such that

$$A \cup B = B, \quad A \cap B \neq B, \quad A \cap C = \varnothing \quad \text{and} \quad B \cap C \neq \varnothing.$$

Draw the sets A and C in the Venn diagram.




(N2011/1/10)

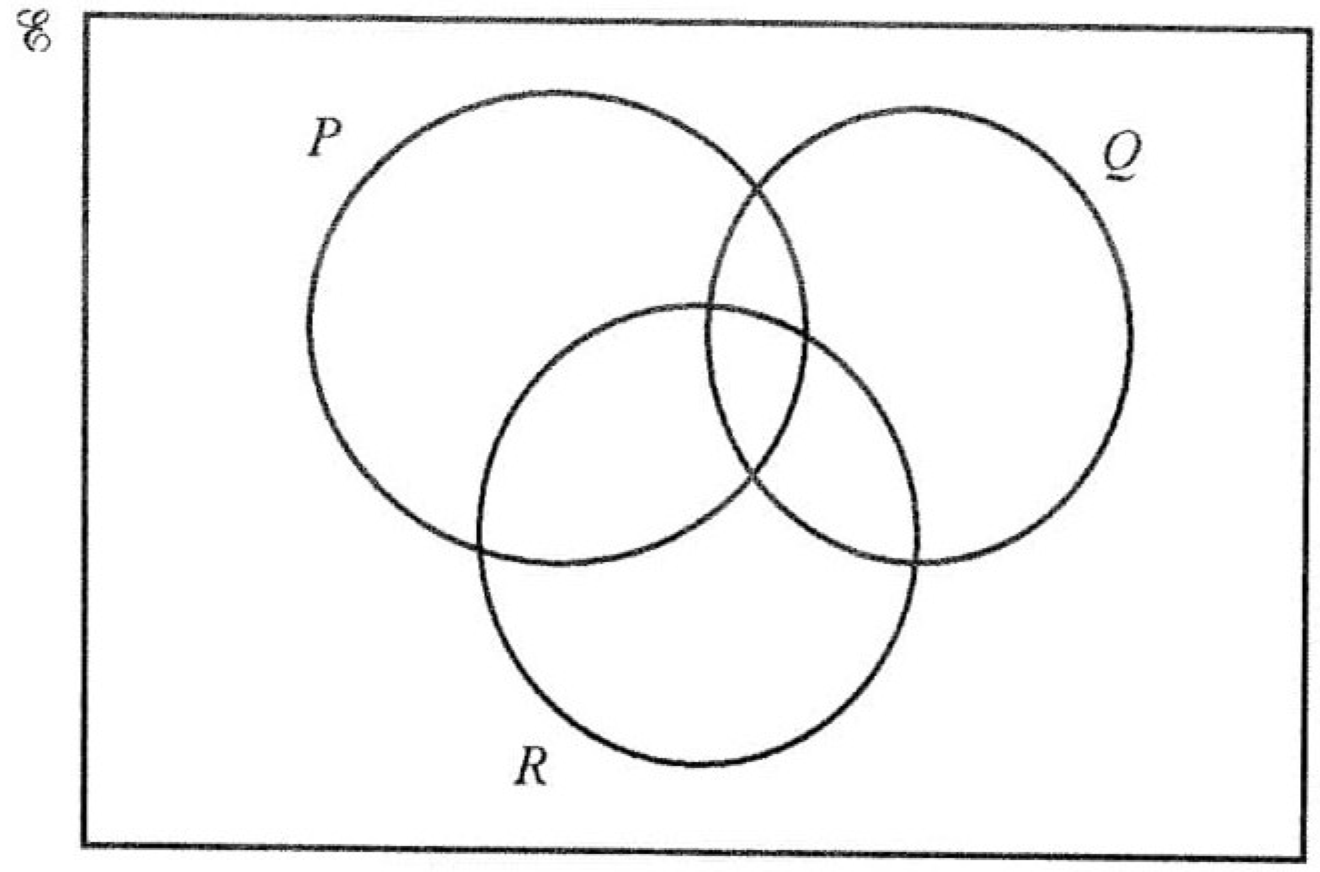
22. (a) $\mathscr{E} = \{ x : x$ is an integer, $2 \leq x \leq 14 \}$
$A = \{ x : x$ is a prime number$\}$
$B = \{ x : x$ is a multiple of 3$\}$

(i) List the members of $(A \cup B)'$

(ii) Find $n(A \cap B)$.

(iii) Given that $C \subset A$, $n(C) = 3$ and $B \cap C = \varnothing$, list the members of a possible set $C$.

(b) On the Venn diagram, shade the set $(P \cup R) \cap Q'$



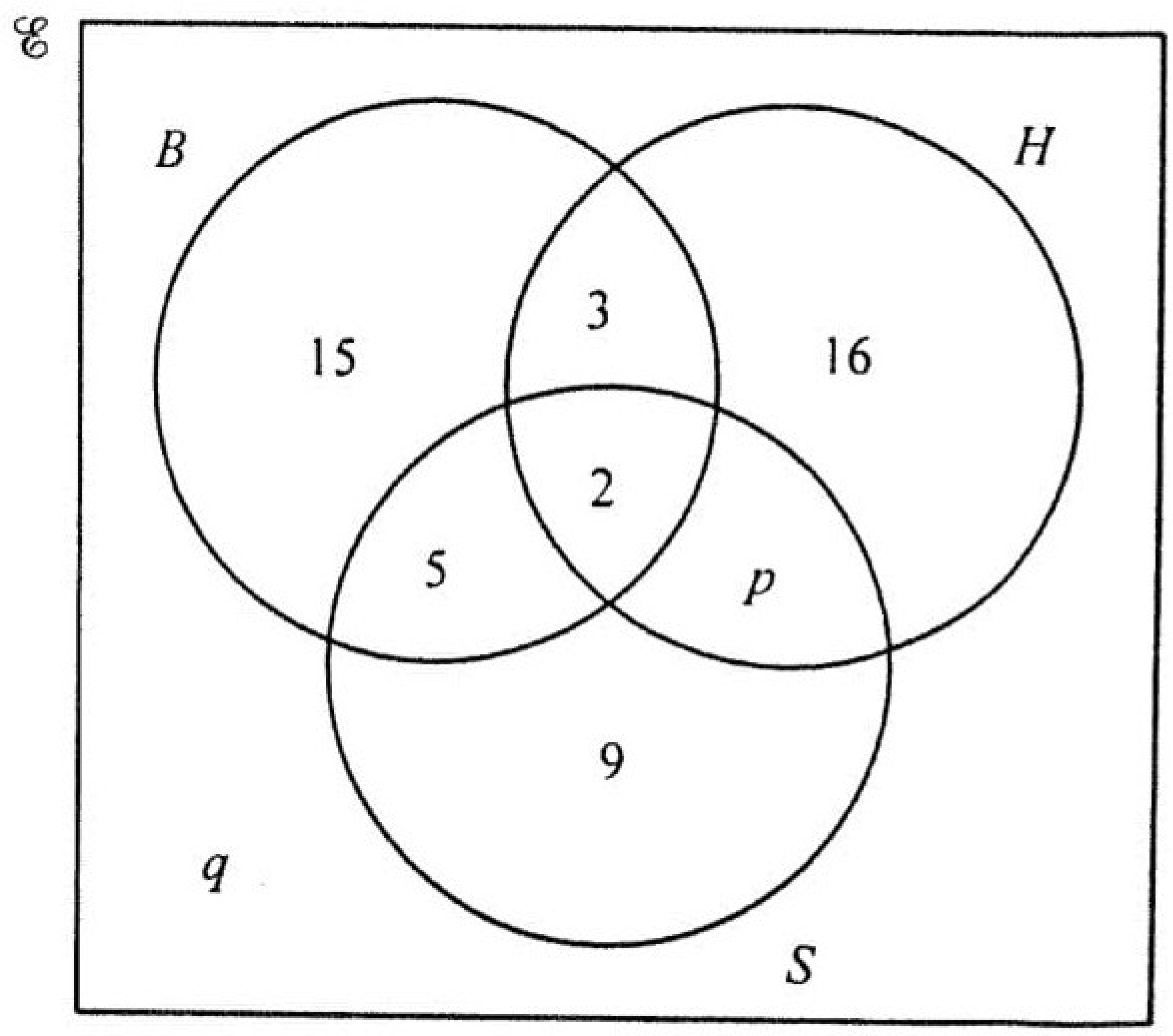

(c) A group of 80 people attended a recreation centre on one day.
Of these people, 48 used the gym
31 used the swimming pool
17 used neither the gym nor the swimming pool.

By drawing a Venn diagram, or otherwise, find the number of people who used both the gym and the swimming pool. (J2012/2/6)

23. In a survey, 60 students are asked which of the subjects Biology ($B$), History ($H$) and Spanish ($S$) they are studying.

The Venn diagram shows the results.

27 students study History.




(a) Find the values of $p$ and $q$.

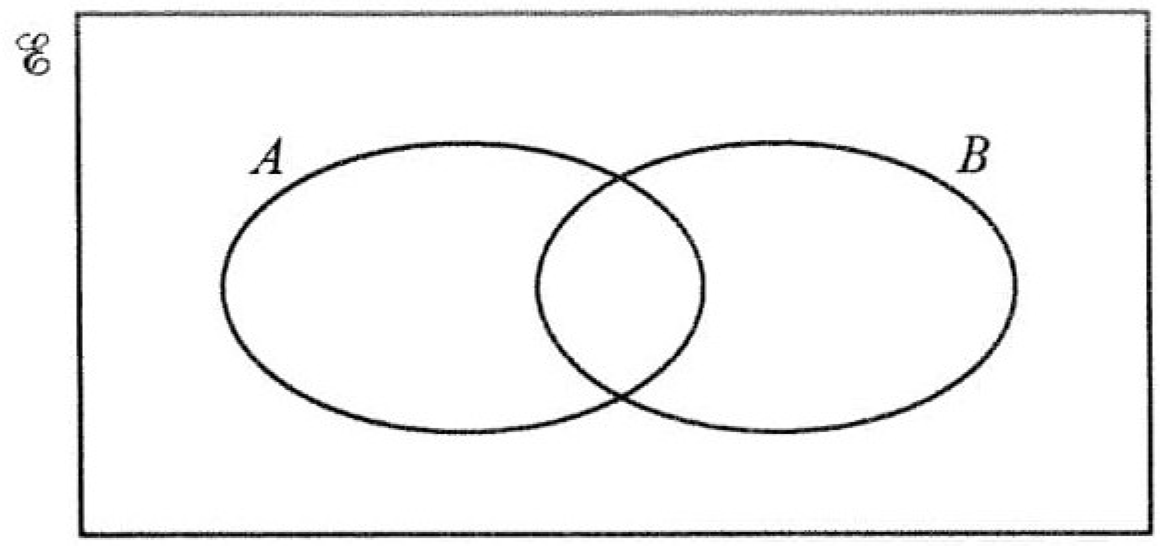
(b) Find $n(H')$.

(c) Find $n((B \cup H) \cap S')$.

(N2012/1/14)

24. $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{odd numbers}\}$
$B = \{\text{multiples of } 3\}$

(a) Complete the Venn diagram to illustrate this information.




(b) Find the value of $n(A \cup B)$.

(c) List the elements of the set $A \cap B'$. (J2013/1/10)

25. (a) $\mathscr{E} = \{x : x \text{ is an integer and } 2 \leq x \leq 12\}$
$M = \{x : x \text{ is a multiple of } 3\}$
$P = \{x : x \text{ is a prime number}\}$

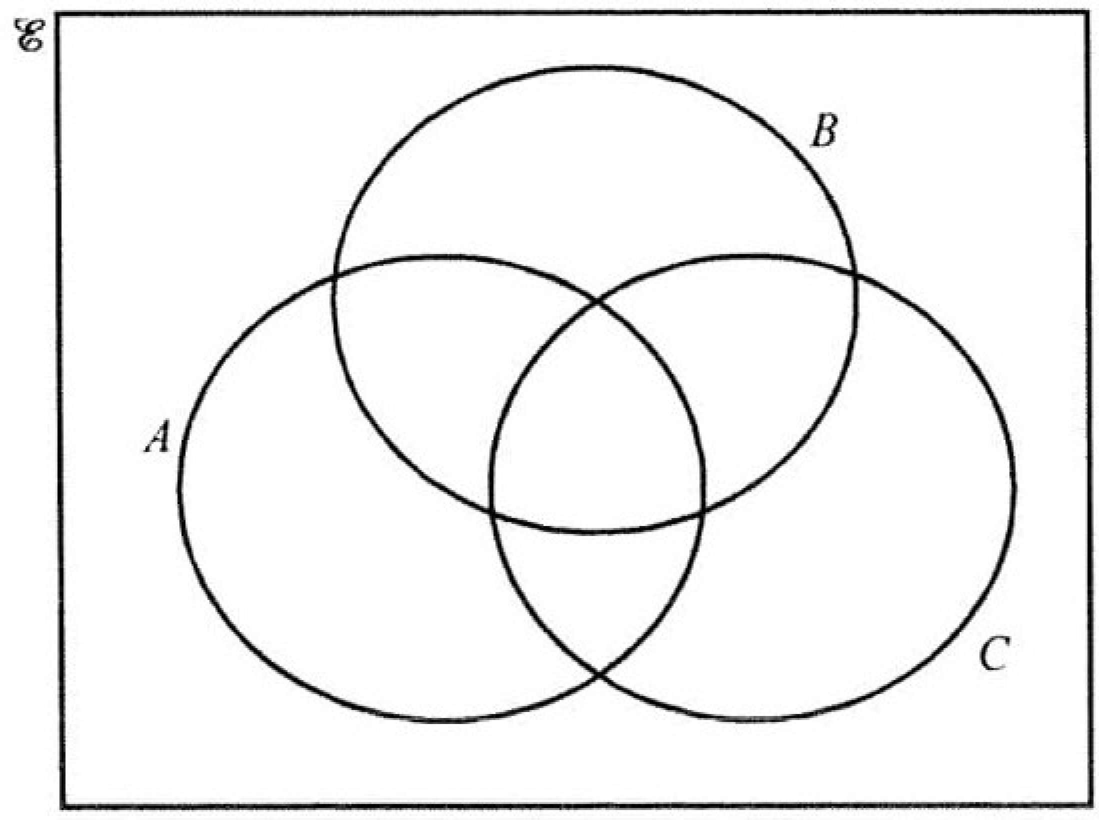
(i) $a \in M \cap P$
Find $a$.

(ii) Find $(M \cup P)'$.

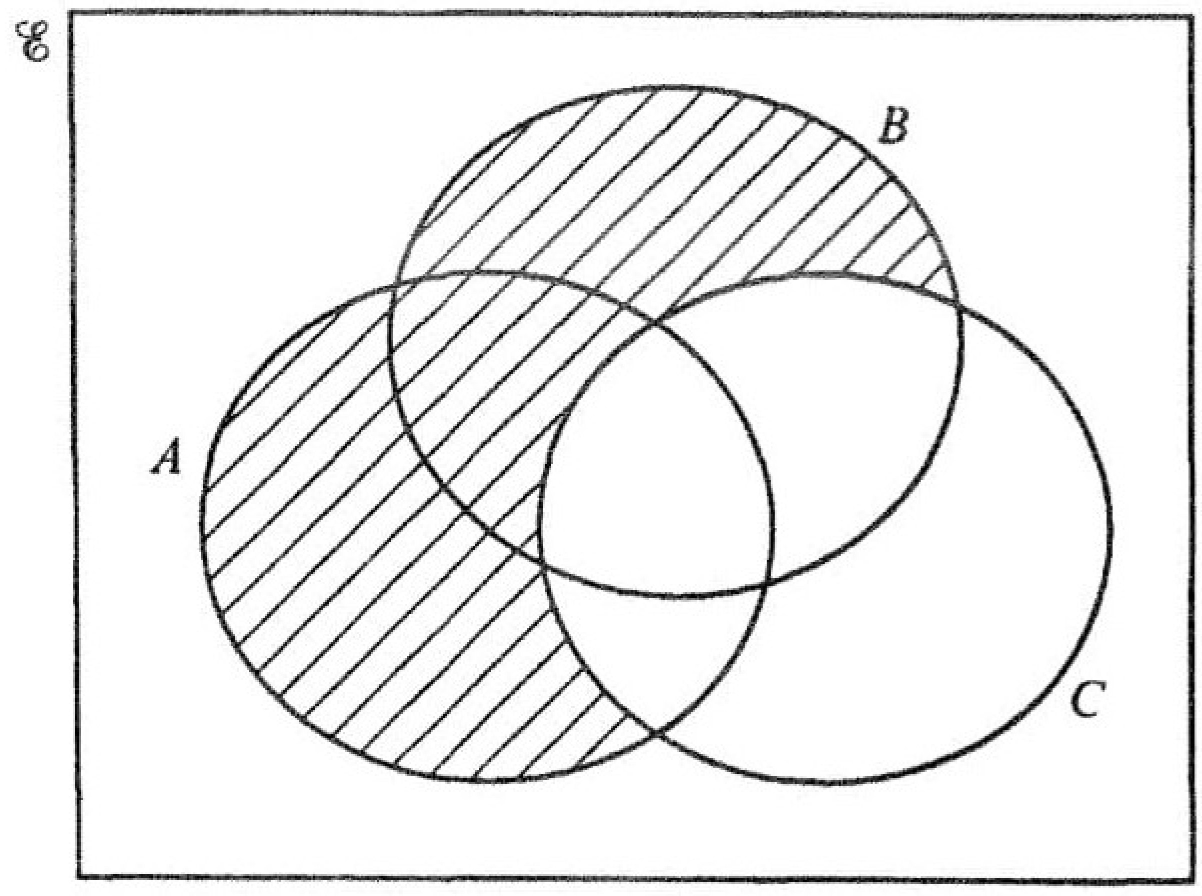
(b) In a survey, 90 people were asked "Do you own a car?" and "Do you own a bicycle?".
A total of 27 people said they owned a bicycle.
Of these, 13 owned **only** a bicycle.
11 people owned neither a car nor a bicycle.
By drawing a Venn diagram, or otherwise, find how many people said that they owned a car.

(c) The Venn diagrams show a Universal set, $\mathscr{E}$, and subsets $A$, $B$ and $C$.

(i) Shade the set $(A \cup C)' \cap B$.

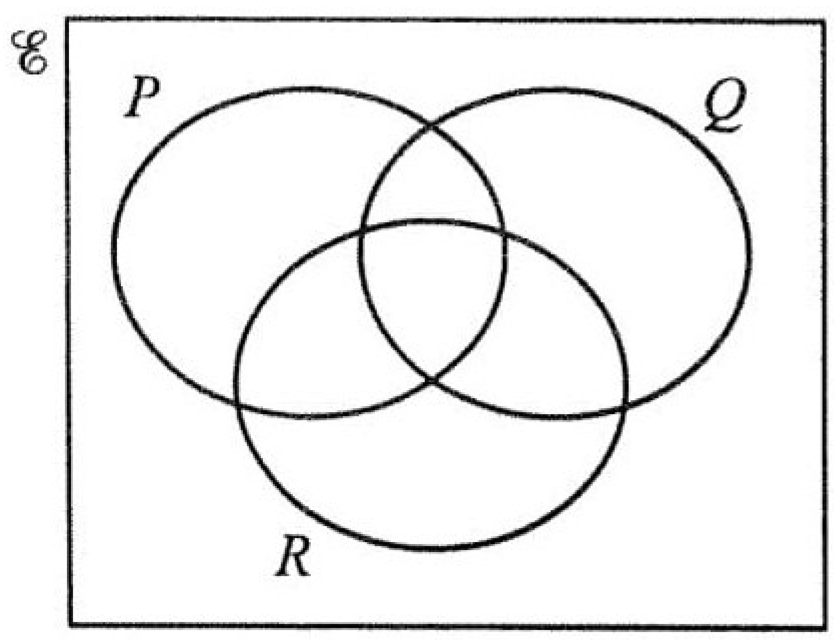


(ii) Express in set notation the subset shaded in this diagram.




(N2013/2/5)

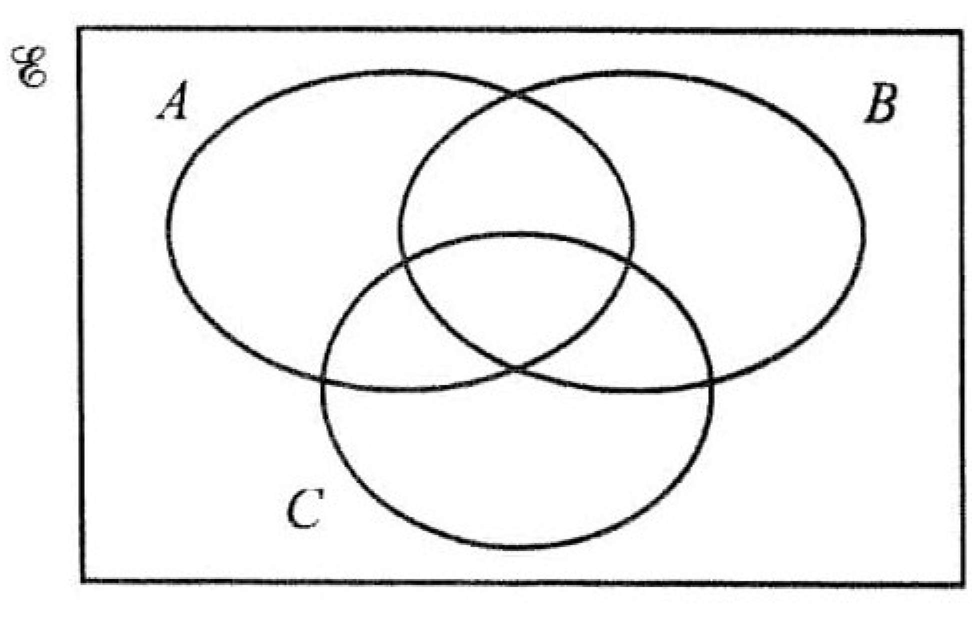
26. (a) On the Venn diagram, shade the set $P' \cap (Q \cup R)$.




(b) A group of 40 children are asked what pets they own.
Of these children, 13 own a cat, 5 own both a cat and a dog and 15 own neither a cat nor a dog.

Using a Venn diagram, or otherwise, find the number of children who own a dog, but not a cat. (J2014/1/11)

27. (a) On the Venn diagram, shade the $C' \cap (A \cup B)$.



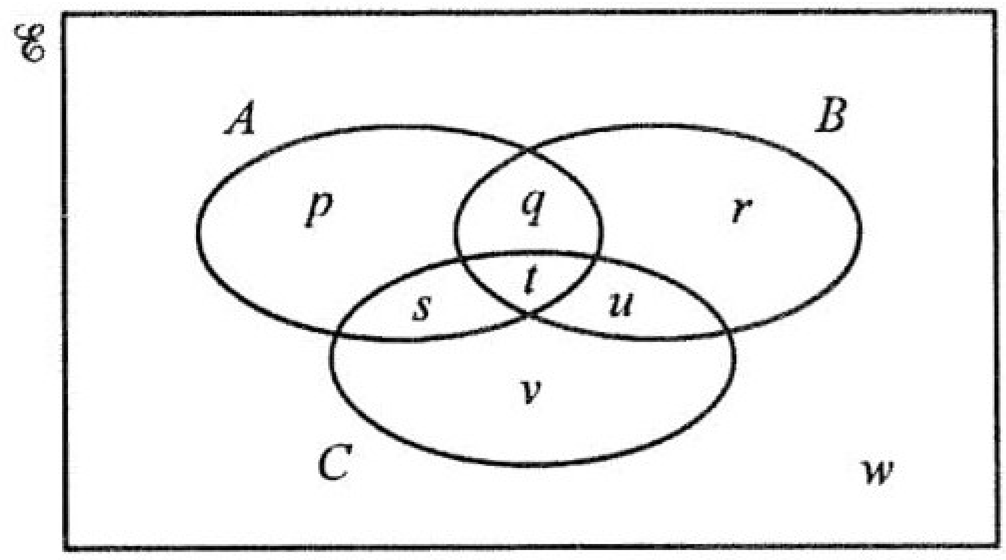
(b) $\mathscr{E} = \{-1, 0, 1, 2, 3, 4, 5, 6\}$

$P = \{-1, 0, 1, 2\}$

$Q = \{x^2 : x \in P\}$

Find $n(Q)$. (N2014/1/6)

28. The Venn diagram shows the sets $A$, $B$ and $C$.

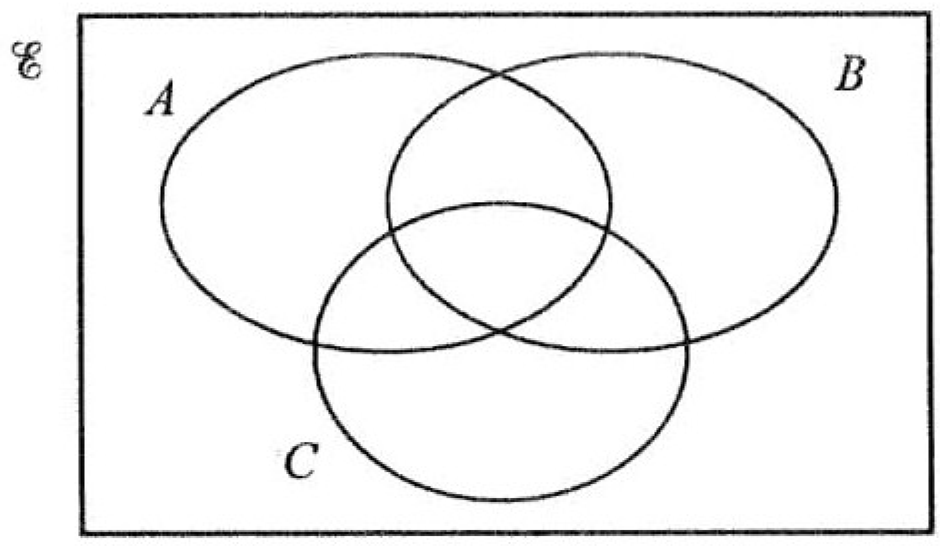



List the elements of

(a) $A \cup B$,

(b) $B' \cap C$. (J2015/1/8)

29. (a) On the Venn diagram, shade the set $B \cap (A \cup C)'$.




(b) $\mathscr{E} = \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19\}$

$W = \{x : x \text{ is a multiple of } 2\}$

$H = \{x : x \text{ is a multiple of } 3\}$

(i) Find $n(W \cup H)$.

(ii) List the members of $W \cap H'$. (N2015/1/15)